CO1- Installation

1. Install the Flutter SDK on your system and configure the required tools for Flutter development. Install Visual Studio Code (or Android Studio), configure the Flutter and Dart extensions, verify the installation using Flutter Doctor, and create and execute your first Flutter application.

2. Design a Flutter application to implement a Simple Calculator. The application should accept two numbers, perform Addition, Subtraction, Multiplication, and Division, and display the result. Also include a Clear button to reset the input fields and the result. Use appropriate Flutter widgets to build the user interface.

CO2-Dart Programming

1. Suppose, you often go to restaurant with friends and you have to split amount of bill. Write a program to calculate split amount of bill.

2. Write a password generator in Dart. Be creative with how you generate passwords - strong passwords have a mix of lowercase letters, uppercase letters, numbers, and symbols. The passwords should be random, generating a new password every time the user asks for a new password. Include your run-time code in a main method.

3. Randomly generate a 4- digit number. Ask the user to guess a 4-digit number. For every digit the user guessed correctly in the correct place, they have a “cow”. For every digit the user guessed correctly in the wrong place is a “bull.” Every time the user makes a guess, tell them how many “cows” and “bulls” they have. Once the user guesses the correct number, the game is over. Keep track of the number of guesses the user makes throughout the game and tell the user at the end.

CO3 & CO4

1. Design a Flutter application that displays a list of products with product images, names, prices, ratings, and an "Add to Cart" button.

2. Design a Flutter application that displays a circular profile image with the user's name and designation. When the user taps the profile image, it should smoothly enlarge, change its border color, and display additional profile details (Email and Phone Number). Tapping the image again should restore the original size and hide the additional details. Include an animated "Follow" button that changes to "Following" when pressed.

3. Create a Flutter application to design a Student Registration Form using the Form widget with validation.

4. Design the UI of an Instagram profile page using Flutter. Display the user's profile picture, username, bio, followers, following, posts count, action buttons, highlights, and a grid of images. Use appropriate Flutter widgets to build the interface.

CO5-Database

1. Build a Flutter application with a user interface containing two textfields(for Name and Roll No) and a button. When the button is pressed, the data from the fields should be saved as a new document in a Firestore collection named ‘students’.